LGQ-GD: A Learned 32-bit Retrieval Key for Stellar Neighbourhoods

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Skyshot · https://skyshot.app

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Abstract

Local star-pattern retrieval can be limited by accidental index collisions even when its geometric descriptor is robust to image noise. We present LGQ-GD, a learned 32-bit key for five-star configurations in the Skyshot Solver. A permutation-invariant network maps normalized geometry and local brightness-dominance information into eight coordinates. Empirical cumulative distribution functions equalize their marginal occupancy; a noise-aware bit allocation and reflected Gray coding convert them into a compact key. An image-conditioned uncertainty head ranks nearby keys for probing, including a second parity branch when the canonical orientation is unstable. The contribution is the integration of these components around selective astronomical retrieval, with the same catalogue records and geometric payload as the preceding handcrafted key. Across 72.6 million catalogue records, collision entropy rises from 21.68 to 24.57 bits. At a matched nominal probe budget, typical simulated fields retain 97.2% of the baseline's per-anchor recall while random retrievals per probe fall from 8.82 to 1.08; a learned sign-bit comparator produces 3.30. In a fixed-binary evaluation, the learned key and its probe policy reduce median paired solver latency by 16.4% and full pipeline latency by 6.1%, with the same 93 correct solutions among 96 reference images. We describe the encoding, training and probing rules, explain the source of the advantage, and delimit the evidence, particularly for very wide fields and probability calibration.

Keywords: learned geometric descriptor; stellar neighbourhood; quantization; Gray code; collision entropy; uncertainty-aware retrieval.

1 The retrieval problem and proposed contribution

A blind astrometric solver must associate detections with catalogue stars before it knows the image's pointing, scale or orientation. A local key reduces this problem to a small number of index lookups. Its usefulness depends on two competing properties: corresponding image and catalogue configurations should yield compatible keys, while unrelated catalogue records should rarely occupy the keys being probed. Improving the first property alone can increase the amount of distracting work passed to hypothesis generation.

LGQ-GD addresses the second property while retaining sufficient recall for redundant, multi-anchor solving. Here *LGQ* denotes a learned Gray-coded quantizer and *GD* denotes the selected geometry-plus-dominance feature set. A configuration contains one anchor A, one reference star R and an unordered set of three neighbours $\{S_1, S_2, S_3\}$. Its key retrieves records that carry absolute geometric information. Subsequent pose voting and catalogue verification decide whether the image has been identified; a key hit alone does not establish a correct world coordinate system (WCS).

The method combines four design choices:

1. A shared neighbour encoder and symmetric aggregation remove the need to assign a stable order to the three neighbours.
2. A learned eight-dimensional representation, marginal equalization and training-time distribution regularization spread records more effectively through a fixed 32-bit address space.
3. A noise-aware allocation of bits gives different coordinates different resolutions. Gray coding makes adjacent quantization bins differ in one bit.
4. Image-dependent boundary costs allocate a small probe budget to plausible alternatives, including uncertainty in parity canonicalization.

The scientific claim is a measured retrieval trade-off, not a new principle of geometric invariance or a universal superiority claim over all astrometric solvers. The relevant experiments compare LGQ-GD with the preceding handcrafted key and a learned sign-bit alternative. End-to-end measurements isolate the replacement of the *key and its probe generator* within the same solver binary [10, 11, 12].

2 Relation to prior work

Geometric indexing followed by independent field verification is established in blind astrometry, notably in Astrometry.net [1]. Learning local stellar representations also has precedents: SIGNN uses graph neural networks for star identification [2]. LGQ-GD instead learns a small, catalogue-independent output space for record retrieval. It has no output class for every catalogue star, and its retrieved records remain subject to geometric verification. No direct performance comparison with SIGNN is claimed.

The sum of shared element embeddings follows the Deep Sets construction [3]. Product quantization provides the broader idea of a compact Cartesian product of separately quantized coordinates or subspaces [4]. LGQ-GD uses *scalar* bins in a learned eight-dimensional representation; it does not use conventional subvector k-means codebooks or asymmetric distance tables. The term product quantizer therefore describes the factorized code structure, not equivalence to a standard nearest-neighbour PQ implementation.

Gray's reflected binary code supplies the one-bit adjacency property [5]. Exploring additional query-dependent keys is related in purpose to multi-probe locality-sensitive hashing [6], but the learned encoder has no established LSH sensitivity bounds. Whitening and distribution regularization have precedents in representation learning [7, 8]. The Gaussian potential used here operates in a CDF cube, includes self-pairs and does not inherit a theorem about uniformity on a hypersphere.

3 From a five-star configuration to an embedding

3.1 Normalized geometry and parity

Let $\ell = \|R - A\| > 0$, $e_x = (R - A)/\ell$ and let e_y be its perpendicular unit vector in the local plane. Define

$$u_j = \frac{(S_j - A) \cdot e_x}{\ell}, \qquad v_j = \frac{(S_j - A) \cdot e_y}{\ell}, \qquad j = 1, 2, 3. \tag{1}$$

Catalogue coordinates are formed in a local tangent projection; image coordinates are measured in pixels. A parity statistic

$$q = \sum_{j=1}^{3} \frac{v_j}{u_j^2 + v_j^2} \tag{2}$$

chooses a canonical reflection: if $q < 0$, replace all v_j by $-v_j$ and retain a flip flag. Configurations with a zero reference length or a neighbour coincident with the anchor are invalid. Image candidate generation also retains the baseline's radial validity gate, allowing a centroid-dependent tolerance when a neighbour is closer than the reference.

Table 1: Relevant distinctions. Literature comparisons describe mechanisms; only the internal v1 and sign-GD alternatives are empirically compared in this paper.

Approach	Encoding or retrieval principle	LGQ-GD distinction
Handcrafted key v1	Quantized log-radii and angles, with explicit order and boundary variants.	Symmetric learned representation; approximately equal-mass marginal bins; uncertainty-ranked fixed-budget probes.
Learned sign-GD	A shared set encoder produces 32 thresholded outputs.	Eight scalar-quantized coordinates reuse bits across multiple ordered boundaries; lower measured collision rate.
Classical PQ [4]	Separately quantized vector subspaces support approximate distance evaluation.	Learned scalar coordinates produce exact lookup keys and geometry-bearing record hits.
Multi-probe LSH [6]	Query-directed exploration of several hash buckets.	Boundary costs derive from predicted CDF residual scales and Gray-bit transitions.
SIGNN [2]	Graph-based learned star identification.	Fixed-size retrieval code, with star identity and pose resolved through index records and verification.

For fixed star identities and exact arithmetic, the normalized coordinates are invariant to translation, rotation and uniform positive scale. Summation in Eq. 2 and in the encoder is invariant to neighbour permutation. Canonicalization also handles reflection away from the ambiguous case $q = 0$. These statements do not imply invariance to arbitrary affine transformations, optical distortion or changes in the detected neighbourhood. Numerical perturbations near a parity or quantization boundary can change a code; probing addresses this instability.

3.2 Geometry, dominance and uncertainty features

For each star, its dominance distance d_i is the distance to the nearest brighter star in the relevant catalogue or detection set. Ratios d_i/ℓ add local crowding and brightness-rank context to normalized geometry. Direct magnitude differences are omitted in the selected model, because photometric transfer across real images was less reliable in the development study. Dominance still depends on brightness order and can change when brighter stars are missing, saturated or measured in another passband.

Put $\rho_j = \frac{1}{2}\log(u_j^2 + v_j^2)$ and $\theta_j = \operatorname{atan2}(v_j, u_j)$ after canonicalization. Table 2 gives the complete selected input. The three uncertainty features feed only the noise head; they do not change the embedding or the base key when geometric and dominance inputs remain fixed.

3.3 Symmetric encoder

For neighbour features $f_j \in \mathbb{R}^{16}$, global features $g \in \mathbb{R}^2$ and noise inputs $n \in \mathbb{R}^3$, the network computes

$$h = T\left(\left[\sum_{j=1}^{3} \phi(f_j), g\right]\right) \in \mathbb{R}^{128}, \tag{3}$$

$$z = W_e h + b_e \in \mathbb{R}^8, \qquad \sigma = 0.002 + \exp\big(W_s[h, n] + b_s\big) \in \mathbb{R}^8_{>0}. \tag{4}$$

The shared map ϕ has widths $16 \to 64 \to 64$ and the trunk T has widths $66 \to 128 \to 128$, with ReLU after each layer. Both output heads are linear before the noise transformation. The network contains 32,424 trainable scalars, including the image-only noise head. The catalogue encoder skips that head.

Table 2: Inputs of the selected GD model. Logs are natural. The centroid scale is $s = \max_i \mathrm{clip}(s_i, 0.05, 2.0)$ pixels over the five stars; e is the anchor's distance to the image edge.

Input group	Count	Definition
Per neighbour	1	ρ_j
	8	$\cos(k\theta_j), \sin(k\theta_j)$ for $k = 1, \ldots, 4$
	6	$\sin(a\pi\rho_j), \cos(a\pi\rho_j)$ for $a \in \{1, 2, 4\}$
	1	$\mathrm{clip}(\log(d_j/\ell), -2, 3)$
Global	2	$\mathrm{clip}(\log(d_A/\ell), -2, 4)$ and $\mathrm{clip}(\log(d_R/\ell), -2, 3)$
Image uncertainty	3	$\mathrm{clip}(\log(s/\ell), -12, 0)$, $\mathbf{1}[d_A \geq e]$, $\mathrm{clip}(\log(e/\ell), -1, 4)$

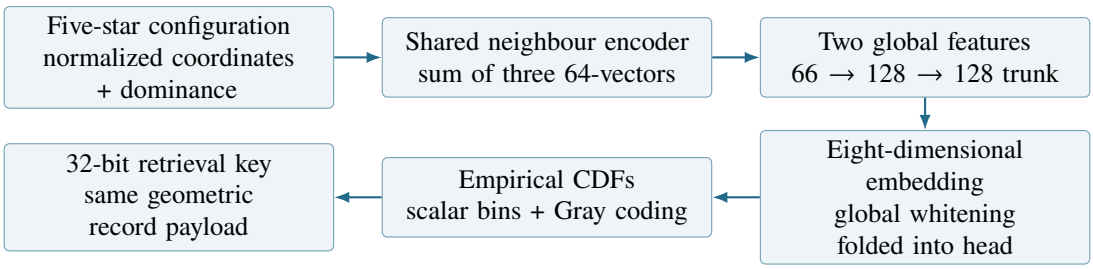


Figure 1: The encoding path. A second reflected branch is evaluated near parity ambiguity. Learned-key hits enter the existing geometric hypothesis and verification stages.

4 Equalized scalar quantization and the 32-bit key

4.1 Empirical CDFs and bit allocation

After training, a global whitening transform is estimated from six million catalogue examples and folded into the embedding head. For each coordinate d, 4,097 stored quantiles define a piecewise-linear empirical cumulative distribution function F_d. The transformed coordinate

$$c_d = F_d(z_d) \in [0, 1] \tag{5}$$

is quantized using b_d bits:

$$v_d = \min\{2^{b_d} - 1, \lfloor 2^{b_d} c_d \rfloor\}, \qquad (b_1, \ldots, b_8) = (4, 3, 3, 3, 4, 5, 5, 5). \tag{6}$$

Uniformly spaced CDF bins are approximately equal-mass on the calibration distribution. Equalized marginals do not guarantee independent coordinates or uniform joint keys. This distinction is central: eight well-balanced coordinates can still occupy only a small correlated subset of the available code space.

The bit allocation uses validation residuals. Starting with one bit per coordinate, each remaining bit is assigned to the coordinate minimizing

$$\Delta_d = 2^{b_d} \widehat{\delta}_d, \tag{7}$$

where $\widehat{\delta}_d$ is the median absolute catalogue–image CDF difference, floored at 10^{-6}. Each coordinate is capped at eight bits. This greedy rule approximates the extra boundary-crossing exposure caused by adding resolution; it is not a proof of globally optimal rate–distortion performance.

4.2 Gray packing and why it helps probing

The reflected Gray code of integer v is

$$\mathrm{Gray}(v) = v \oplus (v \gg 1), \qquad k = \bigvee_{d=1}^{8} \left(\mathrm{Gray}(v_d) \ll \sum_{r<d} b_r \right). \tag{8}$$

Dimension 1 occupies the low bits. For three bits, consecutive bins have codes

$$000,\ 001,\ 011,\ 010,\ 110,\ 111,\ 101,\ 100.$$

A boundary crossing to an adjacent bin therefore flips exactly one bit, whereas ordinary binary encoding can flip several bits at a carry boundary. The converse is false: an arbitrary one-bit flip can reach a nonadjacent bin. The probing procedure uses the distance to actual boundaries, not Hamming distance alone.

Gray coding is a bijective relabelling of bins. By itself it cannot change occupancy, collision entropy or exact-match recall. Those properties come from the learned representation and quantization; Gray coding makes their boundary structure convenient for uncertainty-ranked probing. This separation avoids assigning the observed selectivity gain to the bit labelling alone.

5 Learning stable and well-distributed coordinates

Training pairs contain a catalogue configuration and its simulated image observation. Each minibatch has $B = 8,192$ pairs and $M = 16,384$ additional catalogue examples. The latter support distribution estimation; there is no explicit positive-versus-negative classification loss. Let E stack the $B+M$ catalogue embeddings. Batch whitening uses their mean μ and

$$C = \frac{(E-\mu)^\top (E-\mu)}{B+M-1} + 10^{-5} I = LL^\top, \qquad u = L^{-1}(z-\mu), \tag{9}$$

with column-vector notation for u. The same transform applies to each paired image embedding. During training, $c = \Phi(u)$ uses the standard normal CDF, not the exported empirical table.

The pair term models coordinate residuals $r_{id} = \Phi(u_{id}^{\mathrm{image}}) - \Phi(u_{id}^{\mathrm{cat}})$ with an image-conditioned Laplace scale:

$$\mathcal{L}_{\mathrm{pair}} = \frac{1}{8B} \sum_{i,d} \left(\frac{|r_{id}|}{\sigma_{id}} + \log \sigma_{id} \right). \tag{10}$$

The constant $\log 2$ is omitted. Three distribution terms prevent a stable but uninformative representation:

1. $\mathcal{L}_{\mathrm{marginal}}$ is the mean squared difference between each sorted whitened coordinate and standard-normal quantiles at $(r - \frac{1}{2})/(B+M)$.
2. $\mathcal{L}_{\mathrm{dec}}$ is the mean squared off-diagonal correlation of the 16 standardized features $[u, u^2 - 1]$, using 10^{-6} in the standardization denominator.
3. A Gaussian potential discourages crowded locations in CDF space:

$$\mathcal{L}_{\mathrm{unif}} = \log \left[\frac{1}{U^2} \sum_{i=1}^{U} \sum_{j=1}^{U} \exp\{-200 \|c_i - c_j\|^2\} \right], \qquad U = 8,192. \tag{11}$$

   The selected implementation uses the matched catalogue-side examples for this term and includes diagonal pairs.

The selected objective is

$$\boxed{\mathcal{L} = \mathcal{L}_{\mathrm{pair}} + \mathcal{L}_{\mathrm{marginal}} + \mathcal{L}_{\mathrm{dec}} + 30\mathcal{L}_{\mathrm{unif}}.} \tag{12}$$

Optimization uses AdamW for 10,000 steps, base learning rate 0.002, weight decay 10^{-4}, a 500-step linear warmup with cosine decay, and gradient-norm clipping at 1.0. The selected seed is 0. Its recorded training-loop time was 440.5 s on an RTX 4090, including periodic validation and excluding final export preparation.

Final export recomputes whitening with a $10^{-9}I$ covariance ridge on six million catalogue examples, builds the empirical quantile tables and retains the validation-selected bit allocation. The global transform W is folded into $W'_e = WW_e$ and $b'_e = W(b_e - \mu)$. The noise head is not recalibrated after this change from minibatch normal-CDF coordinates to global empirical-CDF coordinates. Consequently, its scales and derived probabilities should be interpreted as approximate retrieval scores, rather than demonstrated post-export calibrated probabilities.

6 Uncertainty-ranked probing

6.1 Boundary costs and subset search

For bit r of coordinate d, boundaries that flip the Gray bit occur between bins $t-1$ and t with $t = (2m+1)2^r$. Let a_{dr} be the distance in CDF units to the nearest such boundary. The image path computes

$$p_{dr} = \mathrm{clip}\left(\tfrac{1}{2}e^{-a_{dr}/\sigma_d}, 10^{-12}, \tfrac{1}{2} - 10^{-7}\right), \qquad w_{dr} = \log\frac{1-p_{dr}}{p_{dr}}. \tag{13}$$

Flipping a subset J of bits receives cost $\sum_{(d,r)\in J} w_{dr}$. A bit close to an uncertain boundary is inexpensive to change; a distant boundary is expensive. This is a nearest-boundary, factorized approximation. It neither integrates all disjoint intervals sharing a bit value nor models dependencies between bits.

The Rust implementation sorts the 32 costs, retains the 12 smallest and enumerates inexpensive subsets with a best-first heap. It does not materialize all 2^{12} subsets as the offline reference implementation does. Within a retained parity branch, the ordering is exact under this additive surrogate whenever the Kth cost does not exceed the 13th-smallest individual cost. At default $K = 8$, the base key and seven inexpensive singleton flips already establish that the 12 retained bits suffice for this per-branch ranking. This does not establish the most probable keys under the true joint residual distribution.

6.2 Uncertain parity

Near $q = 0$, a small positional perturbation can reverse the canonical reflection. Let σ_q be a first-order propagation of centroid uncertainty through Eq. 2. The alternate branch receives prior cost

$$t = \frac{|q|}{\sqrt{\sigma_q^2 + 0.005^2}}, \qquad w_{\mathrm{parity}} = \log\Phi(t) - \log\Phi(-t). \tag{14}$$

The scale multiplier 1.0 and floor 0.005 were selected on validation pairs from a finite parameter grid. If $w_{\mathrm{parity}} < 12$ nats, the encoder also evaluates the opposite reflection and its probe costs are offset by this prior. The two ranked lists are merged, retaining K candidates.

A production probe is a pair `(key, flip)`. Equal numeric keys with different parity flags remain distinct because they imply different geometric hypotheses. Some offline metric code deduplicates numeric keys, which explains a small difference between the matched nominal and actual probe counts in Section 7.1.

6.3 Index compatibility and compute cost

LGQ-GD retains the same catalogue configurations and 8-byte record payload as the baseline. It changes the key and the interpretation of the parity flag, so an existing v1 index must be re-encoded. The index identifies

its coder; the builder and solver must use the same parameters and quantiles. Retaining the payload format does not make different encoders interchangeable with an already encoded index.

The float32 model blob occupies 260,800 bytes: 129,696 bytes for network parameters and 131,104 for the quantile tables. Metadata adds 2,168 bytes. One uncached image branch requires 42,264 dense multiply–accumulate operations, excluding features, nonlinearities, CDF searches, probes and index access. The catalogue branch requires 41,216. Inference runs on the CPU.

The earlier approximately 1.1 μs microbenchmark combines a dense-kernel measurement with a separate quantizer/log-odds measurement. It excludes important production work and amortizes neighbour embeddings; it is not the complete per-configuration latency. The end-to-end A/B below is the appropriate evidence that reduced retrieval work outweighs the more expensive encoder on the tested system.

7 Evaluation

7.1 Data, baselines and metrics

The offline index contains $N = 72,627,494$ catalogue records. Training data were generated from 20,000 fields in typical and hard scenarios, producing 3,613,474 matched pairs, of which 3,210,419 belong to training and 403,055 to validation, plus six million catalogue samples. Fields span 0.3–60°. A spatial split assigns HEALPix cells [9] at `nside=4` in 70/10/20 proportions; configurations are retained only when all five stars belong to the same split. An anchor is evaluated only if all of its configurations lie in the selected split. The test evaluation contains 154,918 anchors and 10,418,855 configurations. It is spatially separated, but the study was exploratory rather than a pristine preregistered test.

The handcrafted v1 baseline sorts neighbours by log-radius and packs three 5-bit radial fields with one 5-bit and two 6-bit angular fields. It uses 20 radial bins of width 0.12 and angular bins of 6°, with explicit variants near quantization boundaries, neighbour-order ties and parity ambiguity, capped at 96 variants. The sign-GD comparator uses a learned set encoder with 32 thresholded outputs and a learned confidence head. It is a separately trained comparator, not an intervention that isolates Gray coding alone.

An anchor is a true hit if at least one probe retrieves a record with the same anchor and reference-star identities; the other three star identities need not all match. Define

$$a = \frac{H}{A}, \qquad b = \frac{D}{P}, \qquad c = 10^4 \frac{H}{D}, \tag{15}$$

where A counts split-eligible anchors whose true catalogue star has at least one indexed record, and H counts the anchors hit. Probe count P and random retrieval count D aggregate over all split-eligible configurations. For each probe, D counts all catalogue records at that key minus every record belonging to the query anchor. Repeated retrievals across configurations count repeatedly. Thus a is conditional anchor recall, b is distractor work per probe, and c is true-hit anchors per 10,000 random retrieval events. None is a complete-solve probability or runtime speedup.

The learned models receive the number of probes requested by v1 for each configuration, subject to duplicate handling. In the selected LGQ test, the actual totals are 144,935,078 for v1 and 144,926,869 for LGQ, a difference of 0.0057%. We therefore call this a *matched nominal probe budget*.

7.2 Selectivity and occupancy

Table 3 shows that the main gain is selectivity. LGQ-GD retains 97.2% of v1's typical anchor recall and 95.5% of its hard-scenario recall, while producing about eight times fewer random retrievals per probe. Against sign-GD, recall is essentially similar and the collision rate is about three times lower. These are results for the evaluated models; they do not prove an intrinsic capacity limit of all sign-based encoders.

Table 3: Offline simulated test, matched nominal probe budget, all field widths. Efficiency is relative to v1 within each scenario. Values are rounded; ratios use the underlying counts.

Scenario	Key	Anchor recall a	Random hits/probe b	Efficiency c/c_{v1}
Typical	v1	0.501	8.824	1.00
	sign-GD	0.485	3.300	2.59
	LGQ-GD	**0.487**	**1.076**	**7.97**
Hard	v1	0.267	8.308	1.00
	sign-GD	0.254	3.088	2.56
	LGQ-GD	**0.255**	**1.056**	**7.51**

For catalogue counts n_k at code k, collision entropy is

$$H_2 = -\log_2 \sum_k \left(\frac{n_k}{N}\right)^2 . \tag{16}$$

This is the negative logarithm of the collision probability for two independent record draws with replacement. It is not Shannon entropy, and it differs from query-side b because queries follow another distribution and b excludes own-anchor records. Table 4 shows a 2.89-bit increase over v1, corresponding to about $2^{2.89} = 7.42$ times lower empirical record-pair collision probability. For this finite catalogue, empirical entropy is bounded by $\log_2 N \simeq 26.11$ bits, attained if every record has a unique code. The 32-bit bound describes an ideal population distribution over all possible codes.

On the separate offline reference set of 95 frames (88 real and seven synthetic), LGQ-GD retains 0.966 times v1's anchor recall and improves efficiency by 9.54 times. This is not the 96-frame end-to-end set below. On simulated hard fields, efficiency ratios across 0.3–1°, 1–10° and 10–60° are 5.60, 7.28 and 10.61; corresponding typical recall ratios are 0.984, 0.980 and 0.943. A second seed gives lower hard-scenario efficiency ratios, approximately 4.1, 5.4 and 8.7, showing meaningful training variability.

A validation comparison with the same uniformity weight of 30 provides more specific evidence for dominance features. Geometry-only LGQ-G produces 1.314 random hits per probe in both scenarios, compared with 1.064 (typical) and 1.049 (hard) for LGQ-GD. Recall changes only from 0.4810 to 0.4826 and from 0.2500 to 0.2506, respectively. This supports improved selectivity from dominance in the evaluated training pair; several seeds would be needed to quantify the general size of that contribution.

7.3 Fixed probe budgets and wide-field trade-offs

At fixed $K = 1, 4, 8, 64$, typical-scenario anchor recall is respectively 0.383, 0.465, 0.480 and 0.500, versus v1's 0.501 with its adaptive budget. At $K = 8$, hard-scenario recall is 0.248 versus 0.267 for v1. These fixed-budget results must not be confused with the matched-budget results in Table 3.

The reduction is concentrated in difficult geometric regimes. At 40–60°, matched-budget recall ratios are about 0.784 in the typical scenario and 0.724 in the hard scenario; at 20–40° hard, the ratio is about 0.881. In wide-field end-to-end tests, LGQ can require more anchors and the exhaustive pass. Equal final solve counts therefore rely on the tested search policy and available time budget. The evidence supports a selective retrieval design, not uniformly higher recall at every field width.

7.4 Fixed-binary end-to-end comparison

The production comparison uses revision `73106a3`, the same catalogue records, payload, hypothesis voting and verifier, with either v1's adaptive probes or LGQ-GD at $K = 8$. The reference corpus contains 96 images with reference WCS (89 real, seven synthetic). Both variants return 93 correct solutions and no incorrect solutions. Runs use two alternating rounds with three repeats per image, on a shared Apple M5

Table 4: Occupancy of the same 72,627,494 catalogue records. Gray relabelling alone cannot change these values.

Key	Distinct numeric keys	Collision entropy H_2 (bits)
v1	12,290,025	21.6767
sign-GD	24,513,726	22.9282
LGQ-GD	**39,816,459**	**24.5685**

Max system. Each frame's latency is the minimum over the recorded repeats and rounds; it is not cold-start or idle-machine latency.

Table 5 consistently uses the 93 images solved correctly by both variants. The median of paired learned/v1 latency ratios is 0.836 for the solver and 0.939 for the full decode–detect–solve pipeline: reductions of 16.4% and 6.1%. A median paired ratio is not the ratio of the two marginal medians. In the first round's counters, median records read per common image fall from 11,547 to 389, a 29.7-fold ratio of medians; the total pipeline cannot gain by the same factor because detection, decoding, key generation and verification also consume time.

A later check uses 114 file-distinct real frames excluded from model and probe selection. Both variants solve 113, with solutions agreeing within 3 pixels. These frames have no independent reference WCS, so agreement is the claim, not independently established correctness. Of the 114 images, 98 come from the same Crescent Nebula observing session as part of the reference corpus; this is a clustered holdout, not 114 independent sky fields. Median paired ratios are 0.840 for solving and 0.913 end to end. Fresh synthetic checks return 398/400 correct lists for both, 158/160 versus 159/160 correct rendered images, and 73/80 versus 74/80 distorted fields within 3 pixels; both identify all 80 distorted fields.

On 30,000 negative star lists with a 600 ms deadline and eight concurrent jobs, both variants accept zero. V1 reaches the deadline on 11,203 lists; LGQ-GD reaches it on none. Batch wall time falls from 1,122 to 296 s. These measurements come from one fixed-order run and the deadline censors v1's search. Both also accept zero of 200 negative images. Zero observed accepts under these conditions does not establish a zero false-accept probability.

8 Why the method has an advantage, and where it does not

The evidence supports a concrete mechanism. A handcrafted coordinate grid leaves frequent stellar configurations concentrated in relatively few keys. LGQ-GD learns a geometry-dependent transformation and equalizes its marginals; distribution losses address additional correlation. The resulting occupancy substantially reduces the number of unrelated records returned by a probe. The solver can then spend less time constructing and voting on irrelevant hypotheses, while redundancy across anchors compensates for a modest reduction in local recall.

The operational advantages are a compact fixed-size code, a small CPU-resident model, unchanged record payload size, explicit probe-budget control and lower measured distractor work. Symmetric aggregation also removes the encoder's dependence on the order of the three neighbours. The boundary model makes the probing policy responsive to image uncertainty rather than relying only on fixed coordinate tolerances.

Three qualifications delimit these advantages. First, the encoder is more expensive per configuration than v1; gains depend on the collision burden and hardware, and smaller indexes can change the balance. Second, the measured A/B replaces both the key and the probing policy. It does not isolate how much of the benefit comes from the learned geometry, dominance inputs, loss terms, bit allocation or Gray representation. A factorial ablation with several training seeds would be required for that causal attribution. Third, approximate residual scores and marginal equalization do not provide a correctness guarantee. Geometric consensus and WCS verification remain necessary.

Table 5: Reference-image A/B on the common 93 correct solutions. Latencies are in milliseconds; full pipeline includes decoding, detection and solving. Quantiles are recomputed from raw outputs. Record counters use the first round.

Measurement	v1	LGQ-GD, $K = 8$
Solver median	5.009	4.218
Solver 95th percentile	27.055	18.038
Full pipeline median	15.638	14.357
Full pipeline 95th percentile	47.903	40.686
Median records read	11,547	389

The reference data influenced feature selection, particularly the omission of direct magnitudes, and $K = 8$ was chosen using the reference A/B. The original aspiration of 1.5-fold higher per-anchor recall was not achieved. The demonstrated improvement is fewer distracting retrievals and faster complete solving with comparable observed success. Very wide fields remain the principal regression: no real reference image wider than 12.5° was available, and the wider evidence comes from the simulator family used during development. These limits motivate more diverse real-field evaluation and post-export uncertainty calibration.

9 Conclusion

LGQ-GD shows that a learned stellar retrieval key can be valuable without increasing local match recall. Its specific combination of a symmetric geometry-and-dominance encoder, empirical-CDF scalar quantization, noise-aware bit allocation and Gray-boundary probing produces substantially better catalogue selectivity at the same 32-bit width. The measured gain survives the cost of CPU inference and appears as a 6–9% reduction in median paired end-to-end latency in the reported reference and holdout checks. The strongest supported claim is an efficient retrieval component for a verified astrometric pipeline, with clearly identified wide-field and calibration limitations.

A Reproducibility and implementation identity

The selected coder is `models/lgq_gd_u30`. The authoritative implementation consists of `features.py`, `lgq.py`, `probes.py`, `train_lgq.py` and `src/learned.rs`, as listed in the accompanying evidence manifest. Runtime probing follows the Rust implementation where an older prose specification differs, particularly for parity-tagged probes and heap enumeration. The source tree inspected for this manuscript is revision `5f3f3f53`; the timing experiment is the fixed historical revision `73106a3`.

The selected float32 blob has SHA-256

```
1c8649f3ba718db069e8eb9126a2f988
e61a86519d41cc3b6b8859a247e8522a
```

and the coder identity over the blob followed by its JSON metadata is

```
f8fbce6c7c5736f36c60f92431d0e391
8c5d31361a14f4e31536fbe688db88b7.
```

Line breaks are typographic only. The companion `evidence.json` records full paths, hashes, precise metrics and evaluation limitations. Research outputs are local author-maintained artefacts, not an independently archived public benchmark. Reproduction requires the catalogue/configuration data as well as the code and model; the small model alone is insufficient to reproduce the full index and experiments.

B First-order parity uncertainty

For the pre-reflection coordinates, set $r_j^2 = u_j^2 + v_j^2$ and

$$g_{u,j} = -\frac{2u_j v_j}{r_j^4}, \qquad g_{v,j} = \frac{u_j^2 - v_j^2}{r_j^4}. \tag{17}$$

Using the common centroid uncertainty proxy s, the implemented propagation is

$$\sigma_q^2 = \left(\frac{s}{\ell}\right)^2 \left[\sum_j (g_{u,j}^2 + g_{v,j}^2) + \left(\sum_j g_{u,j}\right)^2 + \left(\sum_j g_{v,j}\right)^2 + \left(\sum_j (u_j g_{v,j} - v_j g_{u,j})\right)^2 + q^2 \right]. \tag{18}$$

The expression includes the effects of the shared anchor and reference frame. It is a local positional-noise approximation; missing stars and brightness-order changes are not described by this variance alone.

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